Could Tesla not only survive but also thrive in the automotive industry over the next few years?

I think there are a number of reasons to believe that Tesla can survive and thrive in the automotive industry in the next few years.

First, given how compelling Tesla's pricing and marketing strategy was, it would have positioned the company to thrive. To be specific, Musk's idea of first introducing a high-end luxury product to market the brand as a premium one and then introducing a more affordable version of the product to sell more was phenomenal. He did this by first launching the Roadster in 2012, which retailed for $100,000 and sold 2,500 units. Then he introduced the Model S, which was also a luxury car and whose sales exceeded the combined sales of the BMW 6 and 7 Series and the Mercedes-Benz CLS and S-Class. This was followed by the Model X, which retailed for $81,000, and then finally the Model 3, which cost about $27,500 after the $7,500 federal credit. According to Image 0, this was close to the average price of a car in the U.S. in July 2018. This move attracted 500,000 presale orders, with each order requiring a $1,000 deposit.

Next, although the production stress for Tesla was intense, under Musk's leadership, the company achieved an engineering feat that would have helped it thrive. For example, the Model 3 had 8,000 fewer components than its previous models.Unlike traditional car manufacturers, Tesla produced 80% of its parts in-house, reducing supply chain dependencies. Ford Motors followed this strategy of vertical integration in the 1920s, helping the company thrive in an era when automobiles were a relatively foreign concept. Another factor was that lithium-ion batteries, the most expensive part of a car, accounted for 33% of the final cost and witnessed a sharp drop in price, costing about $100 per kWh by the end of the year for Tesla. On the other hand, GM spent $145 per kWh. There was also a sharp drop in manufacturing time. According to Musk, the company had been able to reduce its battery-pack manufacturing time from 7 hours to less than 17 minutes.

Another reason I believe that Tesla would have been able to thrive is its unique revenue strategies. First, by eliminating middlemen and car dealerships and having its own showrooms (26 in the U.S. and D.C.), Tesla made it really easy for potential customers to buy a car. This saved customers the trouble of going to numerous dealerships to inquire about prices, bargain, and potentially fall for the bad, flashy deals that dealers offered. This was a great move because more millennials were buying their first cars during this time. According to J.D. Power, millennials accounted for 40% of new car owners in 2020. Millennials were known to be a more practical generation that preferred buying cars online to save themselves the trouble of bargaining with employees and to buy a car that was less of a headache because of its low-maintenance nature. For employees, it was a win-win situation, too, because they did not have to depend on commissions and received monthly salaries. Also, according to Exhibit 6a and 6b, vehicle sales accounted for the highest percentage of revenue and profit and Tesla could have benefitted from 100% of it instead of sharing its revenue and profit with middlemen.

Next, Tesla successfully solved “range anxiety” by introducing an extensive network of 1,300 charging stations and 1,100 Superchargers. Musk said he was open to having other car manufacturers share the facilities as long as they shared the cost of usage. The Supercharger offered 170 miles of range for $0.24 per kWh within 30 minutes, compared with the Nissan LEAF, which offered drivers 75–100 miles of range in 30 minutes. As of the May 2018 earnings call, Musk mentioned that a Tesla-based autonomous ride-sharing system was under development and would soon be approved by regulatory bodies. This would benefit Tesla by somewhere between $2 billion and $6 billion in revenue, and a car owner by $6,900 a year. An additional factor that would benefit Tesla was the relatively lower cost of autonomous-vehicle R&D because, while other companies made acquisitions and investments (Uber and Waymo used Toyotas and Fiat Chryslers, and Apple equipped Lexuses), Tesla used its own fleets. As shown in Exhibit 6b, shared mobility was expected to have the largest percentage of profits after vehicle sales.

Furthermore, millennials have identified safety, efficiency, quality, warranty, and comfort as their top five automotive priorities as shown in Exhibit 9, alongside some technology features such as Bluetooth and navigation and Tesla fulfills all those requirements. According to the U.S. News & World Report 2017 luxury car rankings (out of a 10-point scale) as shown in Figure 1, the average vehicle scored 8.8 from critics and 8.7 overall, wherein Tesla Model X earned a 9.1 in both categories, and the Model S achieved a 9.6 critical score and a 9.4 overall score. Consequently, with millennials representing the largest demographic of consumers, Tesla is well-positioned to remain a market favorite and continue to thrive.

Finally, regarding Tesla’s financials, I agree with Musk's belief that "almost any new technology initially has high unit cost before it can be optimized and this is no less true for electric cars." Looking at Tesla’s financial data, Tesla experienced a history of poor financial performance before Q3 of 2018. However, from Q3 of 2018 until Q2 of 2019, it generated positive cash flow except one quarter. Although this sum of positive cashflow is inadequate to make up for negative cash flow prior to Q3 of 2018, I believe if Tesla maintains this strategy of innovative engineering and marketing, it can have an impressive financial performance.

Exhibit 2 Price of Cars in U.S. Market

Segment	July 2018 Transaction Price (Avg.)
Compact Car	$20,412
Compact SUV/Crossover	$28,365
Electric Vehicle	$36,948
Entry-Level Luxury Car	$42,211
Full-Size Car	$35,038
Full-Size Pickup Truck	$48,644
Full-Size SUV/Crossover	$61,557
High-Performance Car	$92,798
High-End Luxury Car	$98,360
Hybrid/Alt. Energy Car	$26,969
Luxury Car	$59,519
Luxury Compact SUV/Crossover	$44,524
Luxury Full-Size SUV/Crossover	$88,038
Luxury Mid-Size SUV/Crossover	$55,118
Mid-Size Car	$25,424
Mid-Size Pickup Truck	$33,125
Mid-Size SUV/Crossover	$37,844
Minivan	$34,635
Sports Car	$31,650
Subcompact Car	$15,026
Subcompact SUV/Crossover	$24,008
Van	$34,902
Average Price	**$35,359**

**Kelley Blue Book* average transaction prices do not include applied consumer incentives.

Exhibit 6a Car Market Value Chain, 2015–2030

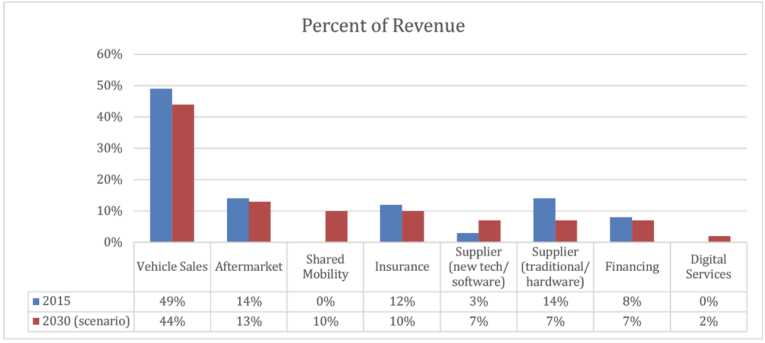
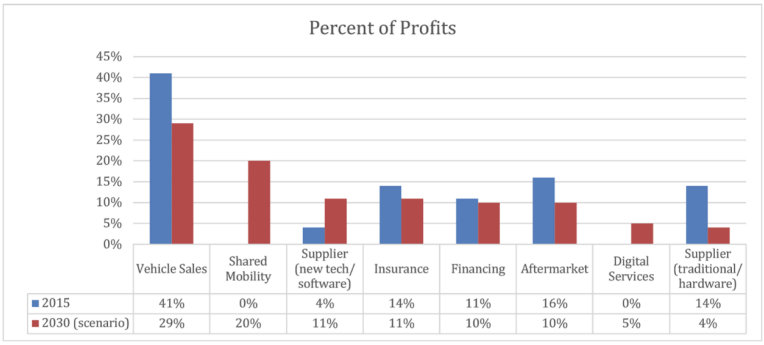


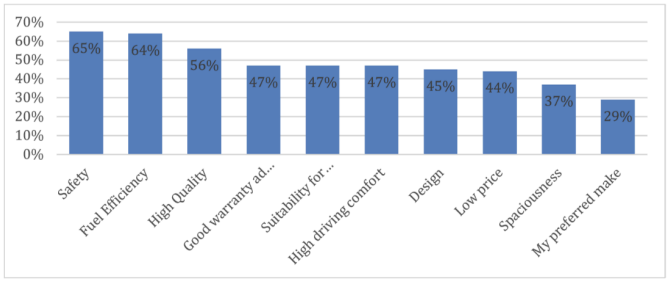
Exhibit 6b Car Market Value Chain, 2015–2030



	Vehicle Sales	Shared Mobility	Supplier (new tech/ software)	Insurance	Financing	Aftermarket	Digital Services	Supplier (traditional/ hardware)
2015	41%	0%	4%	14%	11%	16%	0%	14%
2030 (scenario)	29%	20%	11%	11%	10%	10%	5%	4%

Source: Richard Vierecki, Dietmar Ahlemann, et al. "Connected Car Report 2016," Strategy& (PwC), September 26, 2016.

Exhibit 9 Car-Purchasing Decisions Survey, November 2017–January 2018



Third pary evaluations of luxury cars for 2017 model year

	Owner Satisfaction	Predicted Reliability	Overall Rating	Critics Rating	Overall Score
Maximum possible score	100%	5	100	10	10
Data source	CR	CR	CR	USNWR	USNWR
Tesla Model X	92%			9.1	9.1
Tesla Model S	94%	4	92	9.6	9.4
Audi A7	82%			9.2	9.1
Audi A8	74%		77	8.9	8.5
BMW 6 Series	77%	3		8.2	8.3
BMW 7 Series	86%	4	86	9.1	8.7
Jaguar XJ		2	58	8.1	7.9
Lexus LS	80%	4		8.1	8.3
Mercedes-Benz CLS-Class				8.5	8.4
Mercedes-Benz S	77%	3	78	9.6	9.2
Porsche Panamera 4		3		9.3	9.1
				8.8818182	8.7272727

Source 1: CR = Consumer Reports, a non-profit organization providing reviews of consumer products based on large sample surveys and independent tests of vehicles
Source 2: USNWR = US News and World Report, which bases its rtings on published reviews of cars and reliability data from third parties.
Note: Consumer Reports owner satisfaction = percentage of car owners surveyed who would definitely purchase the same car again

Figure 1 - Third Party Evaluations of Luxury Cars of 2017 Model Year